

Scalability & Future Plan

Date	29 September 2026
Team ID	SWTID-2026-1374
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	READ ACL script uses a fixed branch-based condition	Every new branch or rule needs a manual script change	High
2	Roles bb1 to bb4 are assigned manually to each user	Administration effort grows as the number of users increases	Medium
3	Script is evaluated for every record in the Institution Details table	May slow list views when the record count becomes very large	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Small number of users (EEE test user and administrator)	Assign roles through groups so that new users inherit access automatically
2	Data Storage	Institution Details table with a few student records	Index the branch field and archive old records to keep the table lean
3	Performance	Simple script-based READ ACL	Keep the ACL script lightweight and reuse the branch value from the user profile
4	Security	READ, CREATE, WRITE and DELETE ACLs with roles bb1 to bb4	Add field-level ACLs, audit logs and periodic access reviews

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Group-based role assignment and dynamic branch mapping	Oct - Nov 2026	Less manual work and no script change for new branches
Phase 3	Field-level ACLs with audit and access reports	Dec 2026 - Jan 2027	Stronger security and better traceability
Phase 4	Extend record-level ACLs to other tables and departments	Feb - Mar 2027	Wider coverage of secure record access